

Grouping of Data

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Review of previous session

- ▶ Need of Joining of tables
- ▶ Types of Join
- ▶ Implementation of Cartesian Product, Inner Join, Outer Join and Self Join.

Need of Grouping of Data

To perform aggregate function on a group instead of whole data.

For example:

- ▶ Find total number of students in each class.
- ▶ Find total number of boys and girls students.
- ▶ Find total number of students from each city.
- ▶ Find maximum marks secured by boys and girls student respectively.

In all these cases, we have to perform aggregate function on each group of data instead of all records of the table.

Grouping Data with GROUP BY

- ▶ **GROUP BY** clause is used to group the data on a particular column or group of columns. It divide rows in a table into smaller groups.

Syntax of GROUP By Clause

► Syntax:

```
SELECT column, group_function(column) FROM table  
[WHERE condition]  
[GROUP BY group_by_expression]  
[ORDER BY column];
```

Working of Grouping of data

- ▶ Steps to find total salary of employees belonging to each department:
 - i. Group records on the basis of department number
 - ii. Apply sum function on salary of each group to obtain the required result.

Similarly in SQL, first GROUP BY clause is used to make a group on deptno column and then sum function is used to calculate total salary for each group.

Working of Grouping of data

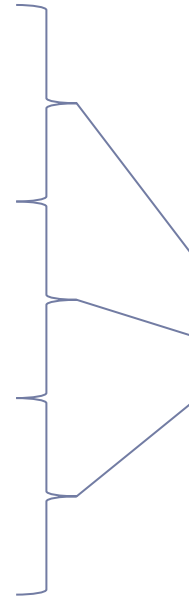
Task-1: Total Salary paid to employees for each Dept Number

Eno	Ename	Job	Salary	Deptno
1	Ram	Asst Prof	40000	10
2	Ramesh	Assoc Prof	60000	10
3	Rahat	Prof	80000	10
4	Suresh	Asst Prof	40000	20
5	Ruhani	Asst Prof	45000	20
6	Rehmat	Assoc Prof	65000	20
7	Ankit	Asst Prof	50000	30
8	Rishan	Assoc Prof	70000	30
9	Amit	Prof	90000	30

Working of Grouping of data

Task-1: Total Salary paid to employees for each Dept Number

Salary	Deptno
40000	10
60000	10
80000	10
40000	20
45000	20
65000	20
50000	30
70000	30
90000	30

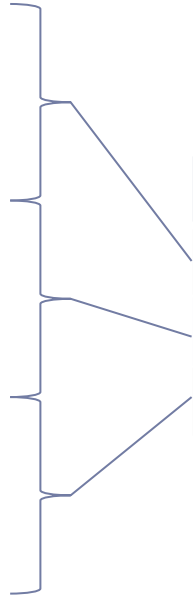


Sum(Salary)	Deptno
180000	10
150000	20
210000	30

Working of Grouping of data

Task-1: Total Salary paid to employees for each Dept Number

Salary	Deptno
40000	10
60000	10
80000	10
40000	20
45000	20
65000	20
50000	30
70000	30
90000	30

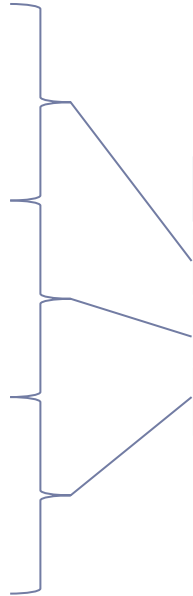


Sum(Salary)	Deptno
180000	10
150000	20
210000	30

Working of Grouping of data

Task-1: Total Salary paid to employees for each Dept Number

Salary	Deptno
40000	10
60000	10
80000	10
40000	20
45000	20
65000	20
50000	30
70000	30
90000	30

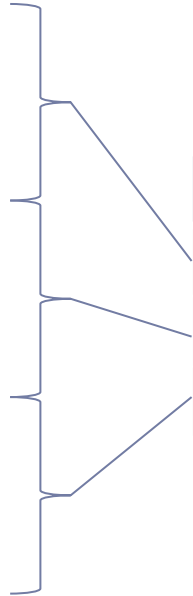


Sum(Salary)	Deptno
180000	10
150000	20
210000	30

Working of Grouping of data

Task-1: Total Salary paid to employees for each Dept Number

Salary	Deptno
40000	10
60000	10
80000	10
40000	20
45000	20
65000	20
50000	30
70000	30
90000	30



Total_Salary	Deptno
180000	10
150000	20
210000	30

Select Sum(Salary) Total_Salary,
Deptno from Employee
Group by Deptno;

Working of Grouping of data

Task-2: Total Salary paid to employees for each Dept Number excluding Prof

Eno	Ename	Job	Salary	Deptno
1	Ram	Asst Prof	40000	10
2	Ramesh	Assoc Prof	60000	10
3	Rahat	Prof	80000	10
4	Suresh	Asst Prof	40000	20
5	Ruhani	Asst Prof	45000	20
6	Rehmat	Assoc Prof	65000	20
7	Ankit	Asst Prof	50000	30
8	Rishan	Assoc Prof	70000	30
9	Amit	Prof	90000	30

Grouping of Data

Working of Grouping of data

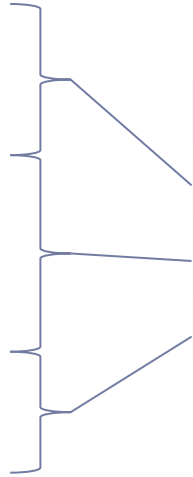
Task-2: Total Salary paid to employees for each Dept Number excluding Prof

Eno	Ename	Job	Salary	Deptno
1	Ram	Asst Prof	40000	10
2	Ramesh	Assoc Prof	60000	10
4	Suresh	Asst Prof	40000	20
5	Ruhani	Asst Prof	45000	20
6	Rehmat	Assoc Prof	65000	20
7	Ankit	Asst Prof	50000	30
8	Rishan	Assoc Prof	70000	30

Working of Grouping of data

Task-2: Total Salary paid to employees for each Dept Number excluding Prof

Salary	Deptno
40000	10
60000	10
40000	20
45000	20
65000	20
50000	30
70000	30



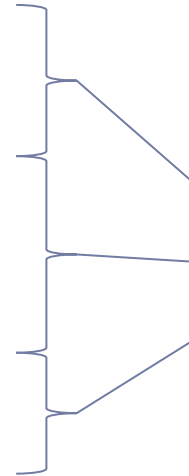
Total_Salary	Deptno
100000	10
150000	20
120000	30

Select Sum(Salary) Total_Salary,
Deptno from Employee
Where Job<>'Prof'
Group by Deptno;

Working of Grouping of data

Task-3: Total Salary paid to employees for each Dept Number excluding Prof arrange the output in ascending order of Total_Salary

Salary	Deptno
40000	10
60000	10
40000	20
45000	20
65000	20
50000	30
70000	30



Total_Salary	Deptno
100000	10
150000	20
120000	30

Working of Grouping of data

Task-3: Total Salary paid to employees for each Dept Number excluding Prof arrange the output in ascending order of Total_Salary

Salary	Deptno
40000	10
60000	10
40000	20
45000	20
65000	20
50000	30
70000	30

Total_Salary	Deptno
100000	10
120000	30
150000	20

```
Select Sum(Salary) Total_Salary,  
Deptno from Employee  
Where Job<>'Prof'  
Group by Deptno  
Order by Total_Salary;
```

Working of Grouping of data

Task-4: Average Salary paid to employees for each Dept Number excluding Prof

Salary	Deptno
40000	10
60000	10
40000	20
45000	20
65000	20
50000	30
70000	30

Avg(Salary)	Deptno
50000	10
60000	30
50000	20

Select Avg(Salary), Deptno
from Employee
Where Job<> 'Prof'
Group by Deptno;

Working of Grouping of data

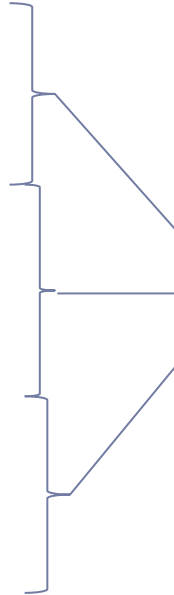
Task-5: Total Number of employees in each Department

Eno	Ename	Job	Salary	Deptno
1	Ram	Asst Prof	40000	10
2	Ramesh	Assoc Prof	60000	10
3	Rahat	Prof	80000	10
4	Suresh	Asst Prof	40000	20
5	Ruhani	Asst Prof	45000	20
6	Rehmat	Assoc Prof	65000	20
7	Ankit	Asst Prof	50000	30
8	Rishan	Assoc Prof	70000	30
9	Amit	Prof	90000	30

Working of Grouping of data

Task-5: Total Number of employees in each Department

Eno	Ename	Job	Salary	Deptno
1	Ram	Asst Prof	40000	10
2	Ramesh	Assoc Prof	60000	10
3	Rahat	Prof	80000	10
4	Suresh	Asst Prof	40000	20
5	Ruhani	Asst Prof	45000	20
6	Rehmat	Assoc Prof	65000	20
7	Ankit	Asst Prof	50000	30
8	Rishan	Assoc Prof	70000	30
9	Amit	Prof	90000	30



Count(*)	Deptno
3	10
3	20
3	30

Select Count(*),
Deptno from Employee
Group by Deptno;

Working of Grouping of data

Task 6: Total Number of employees in each Job

Eno	Ename	Job	Salary	Deptno
1	Ram	Asst Prof	40000	10
2	Ramesh	Assoc Prof	60000	10
3	Rahat	Prof	80000	10
4	Suresh	Asst Prof	40000	20
5	Ruhani	Asst Prof	45000	20
6	Rehmat	Assoc Prof	65000	20
7	Ankit	Asst Prof	50000	30
8	Rishan	Assoc Prof	70000	30
9	Amit	Prof	90000	30

Working of Grouping of data

Task 6: Total Number of employees in each Job

Eno	Ename	Job	Salary	Deptno
1	Ram	Asst Prof	40000	10
4	Suresh	Asst Prof	40000	20
5	Ruhani	Asst Prof	45000	20
7	Ankit	Asst Prof	50000	30
8	Rishan	Assoc Prof	70000	30
2	Ramesh	Assoc Prof	60000	10
6	Rehmat	Assoc Prof	65000	20
9	Amit	Prof	90000	30
3	Rahat	Prof	80000	10

Working of Grouping of data

Task 6: Total Number of employees in each Job

Eno	Ename	Job	Salary	Deptno
1	Ram	Asst Prof	40000	10
4	Suresh	Asst Prof	40000	20
5	Ruhani	Asst Prof	45000	20
7	Ankit	Asst Prof	50000	30
8	Rishan	Assoc Prof	70000	30
2	Ramesh	Assoc Prof	60000	10
6	Rehmat	Assoc Prof	65000	20
9	Amit	Prof	90000	30
3	Rahat	Prof	80000	10



Count(*)	Job
4	Asst Prof
3	Assoc Prof
2	Prof

Select Count(*),
Job from Employee
Group by Job;

Summary

- ▶ Need of Grouping of data
- ▶ Use of GROUP By Clause, WHERE clause and ORDER By of SELECT statement.
- ▶ Next Session
 - ▶ Grouping on more than one column
 - ▶ Some common errors in the use of GROUP By Clause

Grouping on More Than One Column

Objectives

- ▶ Need of grouping on more than column
- ▶ Use of GROUP By clause to group data on more than one column
- ▶ Some common errors in the usage of GROUP By Clause
- ▶ Implementation of GROUP By clause on more than one column

Grouping on more than one column

- ▶ Sometimes there is a need to see results for groups within groups.

For example:

- ▶ To find the total salary being paid to each job title, within each department.

There is a need to having grouping on department number and within each department number grouping on the basis of job or in other words there is a need for grouping within a group.

Thus, EMP table is grouped first by department number, and within that grouping, it is grouped by job title.



Working of Grouping on more than one column

Task 1: Total Salary paid to each Job in each department

Eno	Ename	Job	Salary	Deptno
1	Ram	Asst Prof	40000	10
2	Ramesh	Assoc Prof	60000	10
3	Rahat	Prof	80000	10
4	Suresh	Asst Prof	40000	20
5	Ruhani	Asst Prof	45000	20
6	Rehmat	Assoc Prof	65000	20
7	Ankit	Asst Prof	50000	30
8	Rishan	Assoc Prof	70000	30
9	Amit	Prof	90000	30
10	Vikas	Prof	95000	30

Grouping on more than one column

Working of Grouping on more than one column

Task 1: Total Salary paid to each Job in each department

Deptno	Job	Salary
10	Asst Prof	40000
10	Assoc Prof	60000
10	Prof	80000
20	Asst Prof	40000
20	Asst Prof	45000
20	Assoc Prof	65000
30	Asst Prof	50000
30	Assoc Prof	70000
30	Prof	90000
30	Prof	95000

Deptno	Job	Sum(Salary)
10	Asst Prof	40000
10	Assoc Prof	60000
10	Prof	80000
20	Asst Prof	85000
20	Assoc Prof	65000
30	Asst Prof	50000
30	Assoc Prof	70000
30	Prof	185000

Grouping on more than one column

Working of Grouping on more than one column

Task 1: Total Salary paid to each Job in each department

Deptno	Job	Salary
10	Asst Prof	40000
10	Assoc Prof	60000
10	Prof	80000
20	Asst Prof	40000
20	Asst Prof	45000
20	Assoc Prof	65000
30	Asst Prof	50000
30	Assoc Prof	70000
30	Prof	90000
30	Prof	95000

Deptno	Job	Sum(Salary)
10	Asst Prof	40000
10	Assoc Prof	60000
10	Prof	80000
20	Asst Prof	85000
20	Assoc Prof	65000
30	Asst Prof	50000
30	Assoc Prof	70000
30	Prof	185000

Select Deptno, Job, Sum(salary) from employee
Group By Deptno, Job;

Working of Grouping on more than one column

Task 2: Total Salary paid to each Job in each department excluding Assoc Prof

Eno	Ename	Job	Salary	Deptno	
1	Ram	Asst Prof	40000	10	
2	Ramesh	Assoc Prof	60000	10	✗
3	Rahat	Prof	80000	10	
4	Suresh	Asst Prof	40000	20	
5	Ruhani	Asst Prof	45000	20	
6	Rehmat	Assoc Prof	65000	20	✗
7	Ankit	Asst Prof	50000	30	
8	Rishan	Assoc Prof	70000	30	✗
9	Amit	Prof	90000	30	
10	Vikas	Prof	95000	30	

▶ Grouping on more than one column

Working of Grouping on more than one column

Task 2: Total Salary paid to each Job in each department excluding Assoc Prof

Eno	Ename	Job	Salary	Deptno
1	Ram	Asst Prof	40000	10
3	Rahat	Prof	80000	10
4	Suresh	Asst Prof	40000	20
5	Ruhani	Asst Prof	45000	20
7	Ankit	Asst Prof	50000	30
9	Amit	Prof	90000	30
10	Vikas	Prof	95000	30

Working of Grouping on more than one column

Task 2: Total Salary paid to each Job in each department excluding Assoc Prof

Deptno	Job	Salary
10	Asst Prof	40000
10	Prof	80000
20	Asst Prof	40000
20	Asst Prof	45000
30	Asst Prof	50000
30	Prof	90000
30	Prof	95000

Deptno	Job	Total_Salary
10	Asst Prof	40000
10	Prof	80000
20	Asst Prof	85000
30	Asst Prof	50000
30	Prof	185000

Select deptno, job, Sum(Salary) Total_Salary from employee
where job<>'Assoc Prof' Group By deptno, Job;

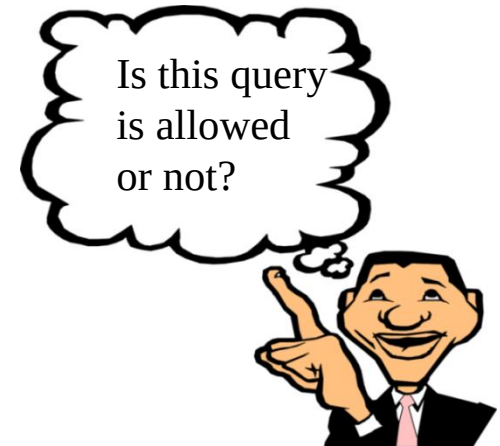
Some common errors

In the usage of GROUP By Clause

- ▶ Whenever we use aggregate or group functions as selected fields in Select statement, then only those columns are allowed in the selected column list that exist in the GROUP By clause of SELECT statement.
- ▶ Example:
- ▶ `Select deptno, count(*) from employee
group by deptno;`
-- It is allowed

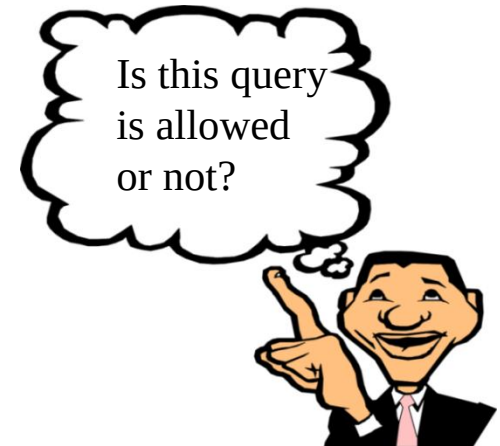
Some common errors In the usage of GROUP By Clause

- ▶ `Select deptno, job, count(*) from employee
group by deptno, job;`
-- Yes, it is allowed.



Some common errors In the usage of GROUP By Clause

- ▶ `Select eno, job, count(*)`
`from employee`
`group by deptno,job;`
`-- illegal use of GROUP By clause`

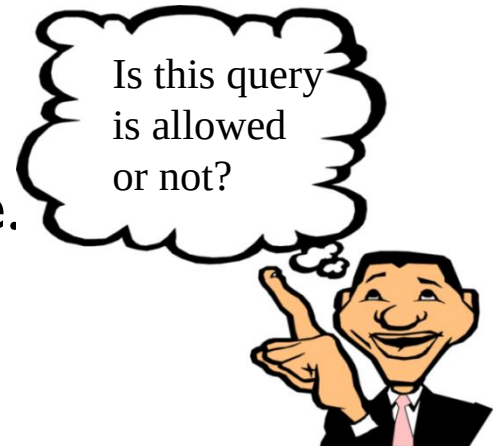


Some common errors In the usage of GROUP By Clause

- ▶ Select deptno, count(*) from employee;
-- illegal use Column missing in the Group By clause

Note:

Any column or expression in the
SELECT list that is not an aggregate
function must be in the GROUP By clause.



Some common errors

In the usage of GROUP By Clause

- ▶ Whenever we use aggregate or group functions as selected fields in Select statement, then only those columns are allowed in the selected column list that exist in the GROUP By clause of SELECT statement.



Summary

- ▶ Need of grouping on more than column
- ▶ Use of GROUP By clause to group data on more than one column
- ▶ Some common errors in the usage of GROUP By Clause
- ▶ Implementation of GROUP By clause on more than one column
- ▶ Next Session:
 - ▶ Use of HAVING Clause of SELECT statement with GROUP By clause

SELECTION OF GROUPS WITH HAVING CLAUSE

Need of Having Clause

- ▶ To select only desired groups that satisfy given condition.
- ▶ For this we have HAVING option of GROUP BY Clause of SELECT statement.
- ▶ It is used to restrict groups based on some condition.
- ▶ As we use WHERE clause to restrict the rows that we select, we have HAVING clause to restrict groups.



Working of HAVING Clause

Total Salary paid to employees for each Dept Number

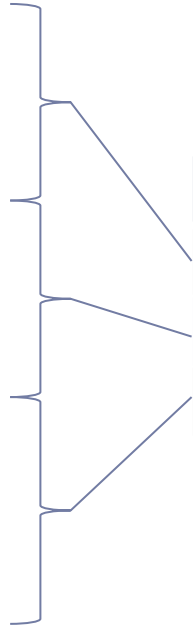
Eno	Ename	Job	Salary	Deptno
1	Ram	Asst Prof	40000	10
2	Ramesh	Assoc Prof	60000	10
3	Rahat	Prof	80000	10
4	Suresh	Asst Prof	40000	20
5	Ruhani	Asst Prof	45000	20
6	Rehmat	Assoc Prof	65000	20
7	Ankit	Asst Prof	50000	30
8	Rishan	Assoc Prof	70000	30
9	Amit	Prof	90000	30

Selection of Groups with Having Clause

Working of Grouping of data

Total Salary paid to employees for each Dept Number

Salary	Deptno
40000	10
60000	10
80000	10
40000	20
45000	20
65000	20
50000	30
70000	30
90000	30

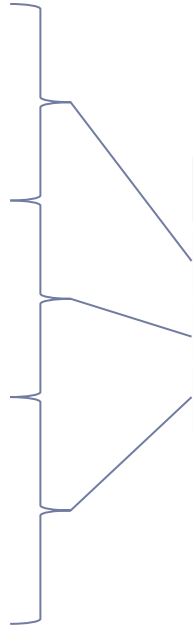


Sum(Salary)	Deptno
180000	10
150000	20
210000	30

Working of Grouping of data

Total Salary paid to employees for each Dept Number

Salary	Deptno
40000	10
60000	10
80000	10
40000	20
45000	20
65000	20
50000	30
70000	30
90000	30



Sum(Salary)	Deptno
180000	10
150000	20
210000	30

Select Sum(Salary), Deptno
from Employee
Group by Deptno;



Working of Grouping of data

To display total Salary paid to employees in the Department and Deptno only for those departments where total salary paid in department is greater than 160000

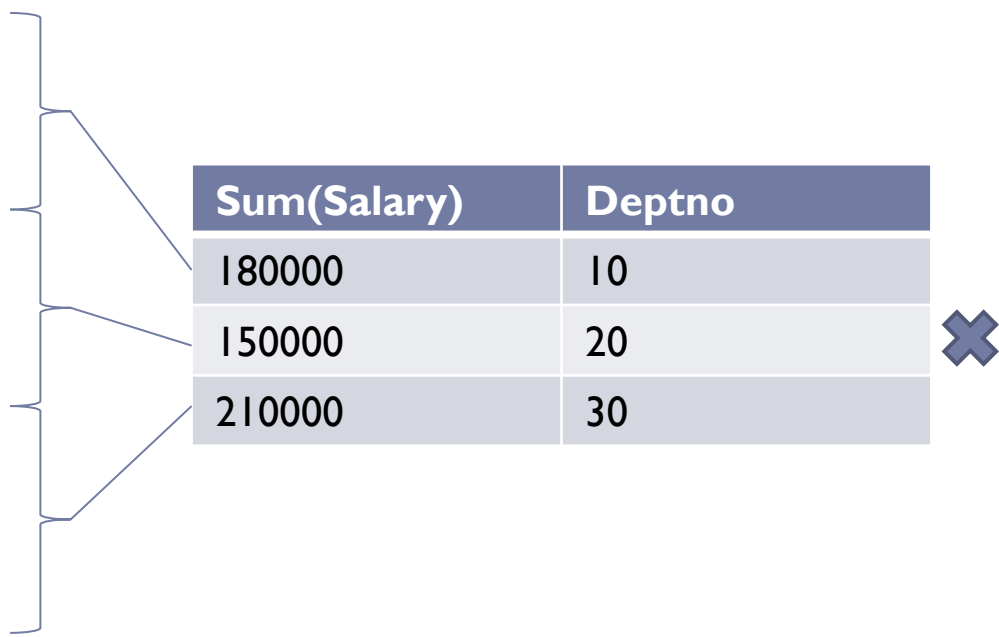
Salary	Deptno
40000	10
60000	10
80000	10
40000	20
45000	20
65000	20
50000	30
70000	30
90000	30

Sum(Salary)	Deptno
180000	10
150000	20
210000	30

Working of Grouping of data

To display total Salary paid to employees in the Department, Deptno only for those departments where total salary paid in department is greater than 160000

Salary	Deptno		Sum(Salary)	Deptno
40000	10	{	180000	10
60000	10		150000	20
80000	10		210000	30
40000	20	{		
45000	20			
65000	20			
50000	30	{		
70000	30			
90000	30			

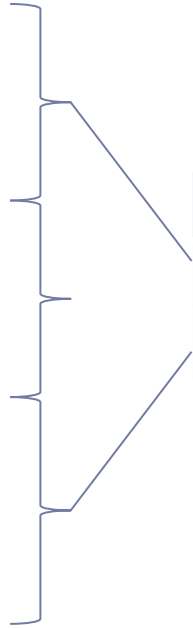


Selection of Groups with Having Clause

Working of Grouping of data

To display total Salary paid to employees in the Department, Deptno only for those departments where total salary paid in department is greater than 160000

Salary	Deptno
40000	10
60000	10
80000	10
40000	20
45000	20
65000	20
50000	30
70000	30
90000	30



Sum(Salary)	Deptno
180000	10
210000	30

Select Sum(Salary), Deptno
from Employee
Group by Deptno
having Sum(Salary) > 160000;



Syntax of HAVING Clause of GROUP By

SELECT column, group_function

FROM table

[WHERE condition]

[GROUP BY group_by_expression]

[HAVING group_condition]

[ORDER BY Column];

It is used to select
desired records

It is used to make
grouping on columns

It is used to select
desired groups

It is used to arrange
O/P in desired order

Order of execution

- i. Rows are grouped
 - ii. The group function is applied to the group.
 - iii. The groups that match the criteria in the HAVING clause are displayed.
- The HAVING clause can precede the GROUP By clause, but it is recommended that you place the GROUP By clause first because it is more logical.



Display department where we pay more than 100000 as total Salary to employees in a single Department excluding Job title Prof. Also display the output in descending order of their department number.

Eno	Ename	Job	Salary	Deptno
1	Ram	Asst Prof	40000	10
2	Ramesh	Assoc Prof	60000	10
3	Rahat	Prof	80000	10
4	Suresh	Asst Prof	40000	20
5	Ruhani	Asst Prof	45000	20
6	Rehmat	Assoc Prof	65000	20
7	Ankit	Asst Prof	50000	30
8	Rishan	Assoc Prof	70000	30
9	Amit	Prof	90000	30

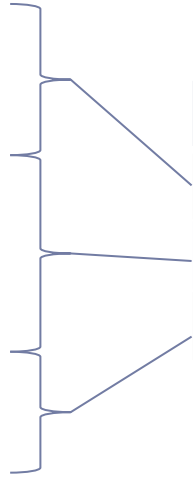


Display department where we pay more than 100000 as total Salary to employees in a single Department excluding Job title Prof. Also display the output in descending order of their department number.

Eno	Ename	Job	Salary	Deptno
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2	Ramesh	Assoc Prof	60000	10
4	Suresh	Asst Prof	40000	20
5	Ruhani	Asst Prof	45000	20
6	Rehmat	Assoc Prof	65000	20
7	Ankit	Asst Prof	50000	30
8	Rishan	Assoc Prof	70000	30

Display department where we pay more than 100000 as total Salary to employees in a single Department excluding Job title Prof. Also display the output in descending order of their department number.

Salary	Deptno
40000	10
60000	10
40000	20
45000	20
65000	20
50000	30
70000	30



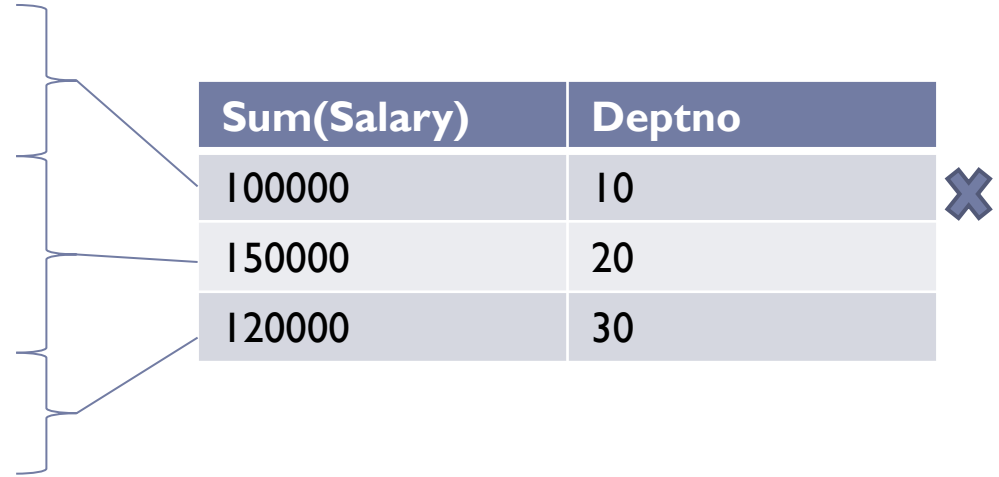
Sum(Salary)	Deptno
100000	10
150000	20
120000	30

Select Sum(Salary), Deptno
from Employee
Where Job<>'Prof'
Group by Deptno;

Display department where we pay more than 100000 as total Salary to employees in a single Department excluding Job title Prof. Also display the output in descending order of their department number.

Salary	Deptno
40000	10
60000	10
40000	20
45000	20
65000	20
50000	30
70000	30

Sum(Salary)	Deptno
100000	10
150000	20
120000	30



Display department where we pay more than 100000 as total Salary to employees in a single Department excluding Job title Prof. Also display the output in descending order of their department number.

Salary	Deptno
40000	10
60000	10
40000	20
45000	20
65000	20
50000	30
70000	30

Sum(Salary)	Deptno
150000	20
120000	30

Select Sum(Salary), Deptno
from Employee
Where Job<>'Prof'
Group by Deptno
having Sum(Salary)>100000;

Display department where we pay more than 100000 as total Salary to employees in a single Department excluding Job title Prof. Also display the output in descending order of their department number.

Salary	Deptno
40000	10
60000	10
40000	20
45000	20
65000	20
50000	30
70000	30

Sum(Salary)	Deptno
120000	30
150000	20

Select Sum(Salary) , Deptno
from Employee
Where Job<>'Prof'
Group by Deptno
having Sum(Salary)>100000
Order by deptno desc;

Some more examples

Example-1

- ▶ Find average salary for each job if more than two employees are working in that job.

```
SQL>SELECT job,avg(salary)
```

```
from employee
```

```
GROUP by job
```

```
HAVING count(*)>2;
```



Example-2

- ▶ List the total, maximum, minimum salary and the average salary of employees job wise, for department number 20 and display only those jobs having average salary greater than 1000

```
SQL>SELECT job, SUM(salary), max(salary), min(salary),  
avg(salary)
```

```
from employee
```

```
WHERE deptno=20
```

```
group by job
```

```
HAVING AVG(salary)>1000
```



Example-3

- ▶ List the total , maximum, minimum salary and the average salary of employees job wise, for department number 20 and display only those rows having average salary greater than 1000 arrange the output in descending order of total salary.

```
SQL>SELECT job, SUM(salary), max(salary), min(salary), avg(salary)
from employee
WHERE deptno=20
group by job
HAVING AVG(salary)>1000
order by 2 desc;
```



Summary

- ▶ Need of Grouping of data
- ▶ GROUP By clause of SELECT statement to group data on a single column
- ▶ Grouping of data in more than one column
- ▶ Need of Selection of groups
- ▶ HAVING clause to select desired groups
- ▶ SELECT statement with WHERE clause, GROUP By clause, HAVING Clause and Order by Clause.

- ▶ Recommended Readings:
 - ▶ More on Grouping of data in reading section of Week-4.



Thanks